

Innovation Outlook

The future impact of new and emerging innovations in the textile value chain

Special Report: Dyes, Inks & Finishes

H2 2026

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Executive summary

The textile & apparel industry's dyeing and finishing sector has been long regarded as one of manufacturing's most resource-intensive stages. Conventional colouration, decoration and finishing processes continue to rely heavily on water, energy and chemical inputs, contributing significantly to industry emissions, wastewater pollution and rising operational costs.

At the same time, increasing environmental scrutiny and tightening regulation means brands, manufacturers and suppliers are now expected to deliver enhanced performance, aesthetics and functionality with a considerably lighter footprint. Ultimately, traditional approaches are proving increasingly incompatible with the demands of a modern textile economy.

The technologies required to transform dyeing and finishing already exist and are maturing. Across the sector, innovators are challenging long-established practices through waterless and low-liquor-ratio dyeing systems, digital and precision colouration technologies, biobased and

non-toxic chemistries, plasma surface treatments, foam-based applications and intelligent process controls designed to minimise waste while improving consistency and productivity.

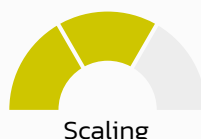
As such, the second Innovation Outlook of 2026 – titled Dyes, Inks and Finishes – explores the products supporting a transition to a greener or more innovative industry. Profiling established technology leaders alongside emerging disruptors, it examines the innovations reducing environmental impact, enhancing process performance and reshaping expectations around textile colouration, decoration and finishing.

In doing so, it aims to provide brands, mills, manufacturers, technologists and investors with valuable insight into one of the industry's most active areas of innovation and the opportunities it presents for long-term sustainable growth.

Otis Robinson,
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Understanding the Commercial Readiness Scale

WTiN's Commercial Readiness Scale gives an indication of what stage of commercialisation a product is at. It ranges from Emerging: a research stage development; Scaling: the product is being produced on a small but growing scale, and Commercially ready: the product is well-established and ready to purchase.



The analysts

Our team of analysts and writers provide coverage on the latest news, detail the newest innovations and explore their expert market analysis across the textile & apparel supply chain.



Jessica Robe

Jessica Robe is WTiN's business innovation lead and sports & outdoorwear specialist. She graduated from De Montfort University with a First Class BA Honours in Contour Fashion, specialising in the design and development of intimates, as well as sports- and swimwear. Since joining WTiN in 2019, Jessica has amassed comprehensive knowledge of the textile & apparel industry, focusing on material and product innovation through the lens of market competitiveness, performance and sustainability.

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Otis Robinson

Otis Robinson is the editor and channel lead of digitalisation at WTiN. Previously, he edited WTiN's *Textile 4.0* and *Knitting International* journals. Prior to joining WTiN, Otis worked in B2B marketing for a variety of Industry 4.0-based clients and freelanced as an investigative and culture journalist. In 2019 he joined WTiN as assistant editor, digitalisation, and has since provided in-depth reportage, features and analysis around textile & apparel industry digitalisation, the application of Industry 4.0 technologies, new socially and environmentally sustainable business models, the industrialisation of manufacturing innovation, and related regulations. He graduated from the University of Hull, UK with a BA in English and Leeds Beckett University, UK with an MSc in Journalism.

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Abigail Turner

Abigail Turner is features editor and podcast host at WTiN. Abigail was previously South West business editor at Reach plc's leading business publication Business Live. She has worked at local and national news publications, primarily writing about women's fashion, lifestyle and culture. Through her work at B2B fashion and beauty news outlets she has provided industry news, insightful analysis, and key data. She joined WTiN as features editor in January 2024 to source and write in-depth features across the entire textile value chain. Abigail is an NCTJ qualified journalist, she graduated from Royal Holloway University of London, UK with a BA in English with Philosophy and London College of Fashion, UK with an MA in Fashion Journalism.

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Dr Yue Guo

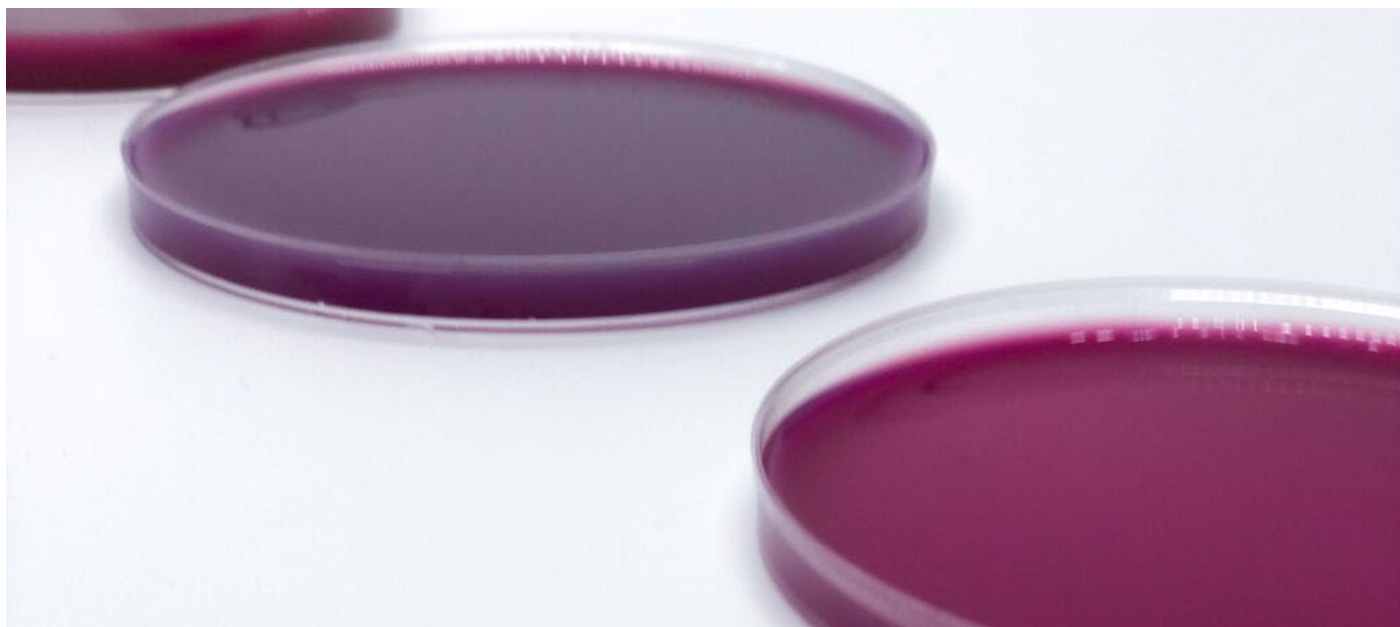
Yue Guo is a textile intelligence analyst at WTiN, specialising in fibres, materials and technology innovation across the global textile and apparel industry. She holds a chemistry PhD focused on conductive ink formulations from the University of Leeds, and a BEng in polymer materials and engineering studied between the Beijing Institute of Fashion Technology and the University of Leeds. At WTiN, Yue works across both editorial and consultancy projects, supporting clients with in-depth analysis of material innovation, sustainability strategies and emerging textile technologies. Prior to joining WTiN, Yue worked as postdoctoral research fellow on multiple industry-collaborative research projects with leading fashion and retail organisations, focusing on garment durability, chemical modification and performances of fibre and fabric.

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Dyes & inks

Exploring notable, innovative products in the dyes and ink sector.





Brilliant Blue, Brilliant Green, Brilliant Red & Brilliant Yellow

Brilliant Dyes



Algae-derived colour has seen great exploration and experimentation in recent years. Akin to many of its peers in the space, Imperial College London spin-out, Brilliant Dyes, makes use of bioreactors to efficiently cultivate microalgae prime for natural colour production.

However, while some have opted to use genetically engineered plants or microbes for their processes, the UK-based startup relies on naturally abundant, colour-concentrated algae species.

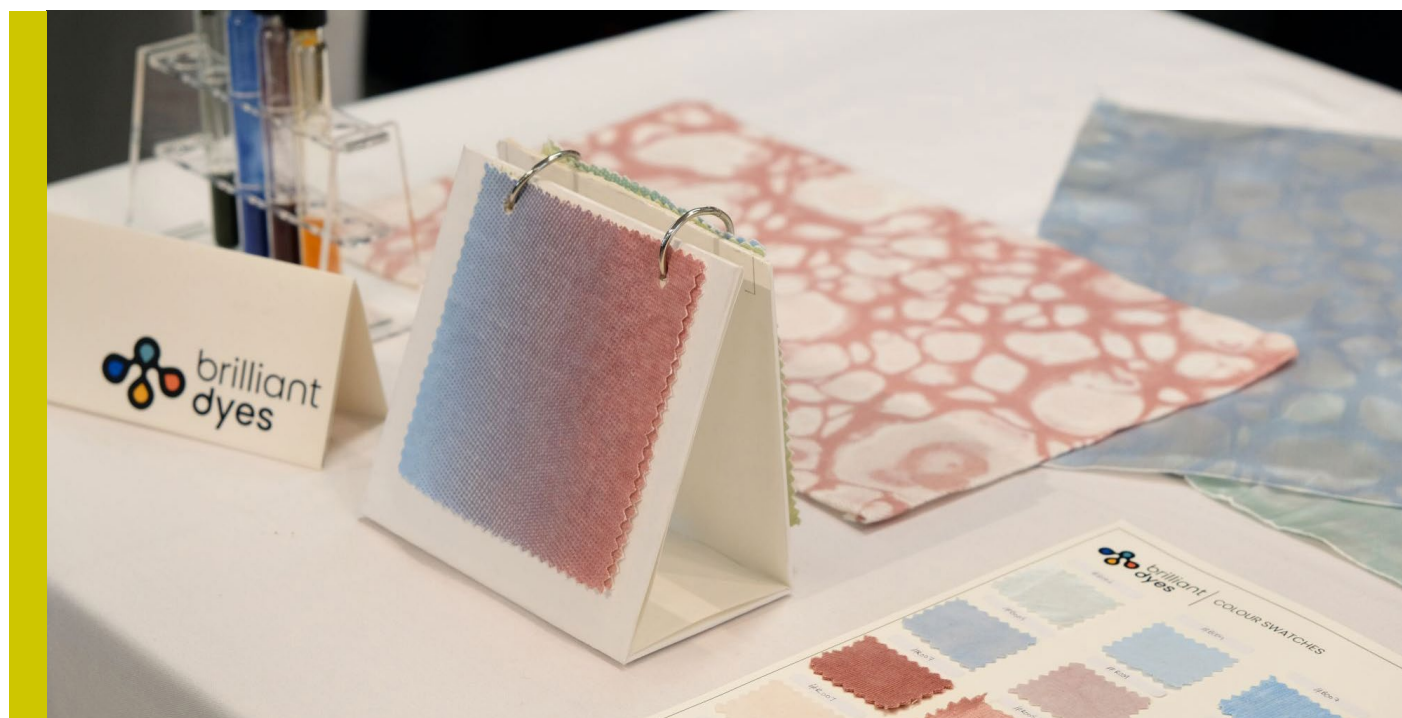
Brilliant Dyes' diverse microalgae library includes those known as cyanobacteria, inclusive of the likes of spirulina – a commonly used dietary supplement. These are optimal

"Naturally abundant algae utilisation gives us the edge of producing natural bioactive colour materials, which are already in the existing safe materials list for even cosmetics and foods."

Dr Mohammad Redwanur Rahman, CEO & co-founder, Brilliant Dyes

for deriving both blue and green dyes. Other algal species such as *dunaliella* and *haematococcus pluvialis* are respectively selected for yellow and red dye production.

From these four organic base shades, the startup uses artificial intelligence (AI) – based models to precisely create a wide spectrum of vibrant colours with differing



The startup uses AI modelling to blend combinations of its four base colours, enabling a width and breadth of derived shades.
Source: Brilliant Dyes



Brilliant Dyes leverages strategically selected microalgae to devise four base colours for dye formulation – blue, green, yellow and red. Source: Brilliant Dyes

intensities – all without use of synthetic ingredients.

Dr Mohammad Redwanur Rahman, CEO & co-founder of Brilliant Dyes, believes this aids scalability, suppresses costs and improves viability across a wider pool of end use industries.

“Naturally abundant algae utilisation gives us the edge in producing natural bioactive colour materials, which are already in the existing safe materials lists for even cosmetics and foods,” Dr Rahman explains. “It gives us the opportunity to scale globally, even in open systems, without the need for class 2 or 3 biosafety labs or closed, confined, expensive reactors, which are necessary for genetically modified organisms (GMOs).”

He continues: “GMOs often require specific chemical or physical stressors – like targeted toxins, heavy metals or precise chemical switches – to force the foreign gene to express the colour, are prevented by strict regulations and have less consumer

acceptance, whereas our natural algae-based solution doesn’t face any of those challenges.”

Once these strategically selected microalgae are ready to harvest, their cell wall integrity is compromised – a low energy process, aided through use of strategic solvents – enabling the extraction of pigment. Once purified and subsequently formulated, these reportedly high-quality biobased dyes are prime for an array of applications, including textiles, cosmetics and food.

Sustainability claims

Brilliant Dyes claims significant impact savings over conventional textile dyes, including 90% less water usage and a 70% lower carbon footprint. Additionally, it reports a 75% lower energy usage during dye extraction versus traditional production, with a 10 tonne reduction in CO₂ per tonne of natural dyes produced.

Dr Rahman points to the carbon neutrality of the process in helping achieve these reductions: “We fix CO₂ in the algae production process, making the whole product carbon neutral and negative at scale.”

The startup’s algal process also claims competitive commercial advantages, with improved margins and lower operational costs over traditional dye production. The profitability of the proprietary platform is aided by the valorisation of the process’ leftover biomass, which – rich in protein and essential amino acids – can be diverted to secondary markets such as organic animal feed and biostimulants. Scaling and validating these derivative products is another objective for the team, providing multiple channels for commercial success.

Furthermore, Brilliant Dyes’ resulting colourants are claimed 100% biodegradable, easing end-of-life management concerns

when paired with likewise materials, components and finishes.

Verification & modelling

To fortify these claims, the team is conducting various experimental works and simulations to assure biodegradation, energy reduction and carbon footprint savings. A dedicated digital twin model assists with this process, enabling accurate emulation of its current technology platform.

Continuous improvement of its data flows, simulation and automation appear set into the business' growth plan. "Our deep tech innovation will undergo step-by-step increases in scale, de-risking and validation – both in-house and third-party." Dr Rahman explains. "We are also gradually automating our platform, which will give us real-time data recording at scale. This will be used for the impact assessment, reporting and validation by experts."

This no doubt will be critical moving forward as key textile markets such as the EU face much more stringent regulation of sustainability claims.

Technology readiness

Currently, Brilliant Dye's algal extraction platform is operating at a TRL-5 (technology readiness level), achieving pre-pilot production levels. Its current system involves a 120 litre, small-scale bioreactor for algae cultivation, paired with a 10-litre continuous, close-loop extraction platform. This allows the team to process 1-2 kg of biomass per day, reportedly equating to a daily dye output of 200 g.

Key to the business' alignment with the textile market, Brilliant Dyes is simultaneously working on the textile fixing performance of its novel algal dyes. So far, this work sits at a TRL-4.

Nevertheless, the ambition is to develop a drop-in solution for the industry, enabling smoother

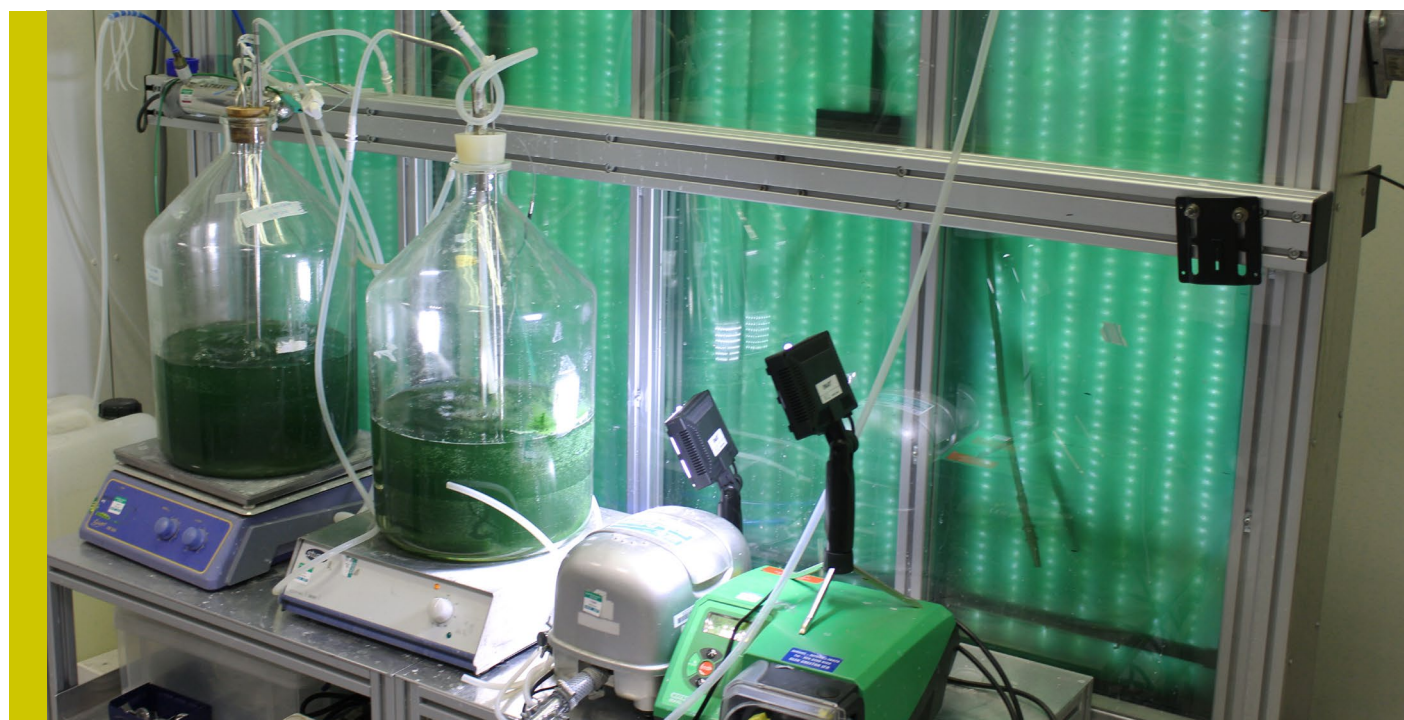
commercial-scale adoption. As part of this, Brilliant Dyes is targeting industrial and academic collaborators to validate its technology against real-world methodologies, setups and textile substrates.

"The industry works with many different textile materials and dyeing and fixation processes," explains Dr Rahman. "We are carefully de-risking and commercially validating with industry partners for each of our bioactive dyes, with each textile material and fixation process."

This de-risking work involves several priorities, including ensuring adequate IP protection and thorough pre-pilot validation. Again, Brilliant Dyes' use of a digital twin is hoped to aid automation and agility in these areas.

Academic roots

The group's current technology platform is the product of numerous academic institutions and collaborations over four and a half years of R&D. Spun out from Imperial (UK) by Professor Klaus

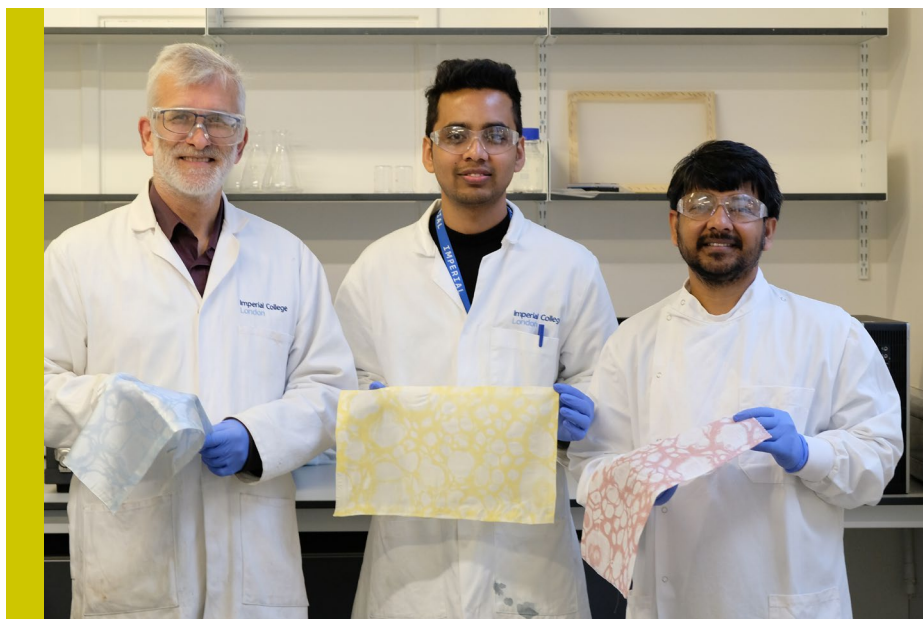


Brilliant Dyes believes its algal production setup is cost competitive and scalable, aided by its use of naturally abundant, industrially safe and colour-rich microalgae. Source: Brilliant Dyes

Did you know?

Imperial College London, UK, has a strong spin-out and startup culture across a number of disciplines, often demonstrating deep tech or biotech-driven solutions. This is inclusive of textiles & apparel, with Brilliant Dyes representing the sector alongside the likes of fellow Imperial spinouts, Dyecycle and Solena Materials.

The first, Dyecycle, is scaling a proprietary technology developed to extract fibres cleansed and void of dyes, coatings and other contaminants from textile waste, creating a valorised, clean waste stream. Elsewhere, Solena Materials leverages the power of AI to engineer and iterate new protein-based materials which could one day replace petrochemical synthetics.



Brilliant Dyes has been spun-out from Imperial College London (UK) by Professor Klaus Hellgardt, Dr Md Tabish Noori and Dr Mohammad Redwanur Rahman. Source: Brilliant Dyes

Hellgardt, Dr Md Tabish Noori and Dr Rahman, the team works from within the university's startup ecosystem; its lab is situated at Imperials' South Kensington campus with additional office space secured at Scale Space, offering dedicated startup support, facilities and services.

A partnership with Scotland's Heriot-Watt University has also been established to further the business' algae dye fixation work – a necessity for progressing the industrial viability of the technology within the textile sector.

Fiscal & technical backing

Key to many startups, funding has been critical to sustaining the business' developmental and commercialisation work. So far, Brilliant Dyes has secured over GB£550k (approximately US\$733k) in non-equity grants, alongside five letters of intent (LoI). The technology's versatility across numerous industries – including textiles, food and cosmetics – has also help it attract diversified interest from over 50 global brands.

Said attention may be pivotal to securing future finance, with Brilliant Dyes planning a pre-seed funding round set to open Q4 2026. The finance raised is positioned to scale production to a pilot-stage capacity by Q3 2027, hoped capable of producing 5-10 tons of dye per year.

For more information, visit www.brilliantdyes.com.

Product	Company	Function	Application(s)	Location
Brilliant Blue, Brilliant Green, Brilliant Red & Brilliant Yellow	Brilliant Dyes	Seaweed-derived, biodegradable dyes	Primarily natural fibres: silk, cellulose, cotton and wool	London, UK

